

Theorem of Gell-Mann and Low

1. Starting Point

- time-ordered Green's function in the Heisenberg picture:

$$G(1, 1') = -\frac{i}{\hbar} \langle T [\psi_H(1) \psi_H^\dagger(1')] \rangle \quad (1)$$

The time dependence of $\psi_H, \psi_H^\dagger$ is governed by the full Hamiltonian $H = H_0 + H_{int}$.

- time-ordered Green's function with field operators in the interaction picture:

$$G(1, 1') = -\frac{i}{\hbar} \langle S(t_0, \infty) T [S(\infty, t_0) \psi_0(1) \psi_0^\dagger(1')] \rangle \quad (2)$$

The time evolution of $\psi_0, \psi_0^\dagger$ determined by H_0 , the influence of H_{int} is described by S-operators.

This is the starting point for expanding in powers of H_{int} .

2. System in the Ground State: $T = 0$

- calculation of expectation values:

$$\langle \dots \rangle = \frac{\langle \phi_G | \dots | \phi_G \rangle}{\langle \phi_G | \phi_G \rangle} \quad (3)$$

- $|\phi_G\rangle$... Heisenberg-picture ground state of the interacting system satisfying

$$H |\phi_G\rangle = E |\phi_G\rangle \text{ with } H = H_0 + H_{int} \quad (4)$$

i.e. $|\phi_G\rangle$ contains the full interaction of the system

- time-ordered Green's function:

$$G(1, 1') = -\frac{i}{\hbar} \frac{\langle \phi_G | S(t_0, \infty) T [S(\infty, t_0) \psi_0(1) \psi_0^\dagger(1')] | \phi_G \rangle}{\langle \phi_G | \phi_G \rangle} \quad (5)$$

again calculated in the Heisenberg-picture interacting ground state $|\phi_G\rangle$ as only the field operators $\psi, \psi^\dagger$ have been rewritten

-  The interaction contained in $|\phi_G\rangle$ needs to be included in the desired expansion in powers of H_{int} .

3. A Trick: Adiabatic Switching of the Interaction

$$H_{int} \rightarrow H_{int} e^{-\varepsilon|t|} \text{ with } \varepsilon \rightarrow 0 \quad (6)$$

The interaction is gradually switched on $t = -\infty$ to finite times t_0 and again slowly switched off from finite times to $t = +\infty$. For $\varepsilon \rightarrow 0$, the properties for finite times are unchanged.

As a result, for $t = \mp\infty$, the system is in the ground-state without interaction:

$$H_0 |\phi_G^0\rangle = E |\phi_G^0\rangle \quad (7)$$

The transition of the system due to the adiabatic switching can be described via

$$|\phi_G\rangle = S_\varepsilon(t_0, -\infty) |\phi_G^0\rangle \quad (8)$$

Here the S-operator provide the evolution of the state from $t = -\infty$ to t_0 under the influence of the slowly ramped up interaction H_{int} . The subscript ε on the S-operator indicates the used adiabatic switching.

4. Gell-Mann and Low Theorem

$$\lim_{\varepsilon \rightarrow 0} \frac{S_\varepsilon(t_0, -\infty) |\phi_G^0\rangle}{\langle \phi_G^0 | S_\varepsilon(t_0, -\infty) | \phi_G^0 \rangle} = \frac{|\phi_G\rangle}{\langle \phi_G^0 | \phi_G \rangle} \quad (9)$$

is an eigenstate of H

$$H \frac{|\phi_G\rangle}{\langle \phi_G^0 | \phi_G \rangle} = E \frac{|\phi_G\rangle}{\langle \phi_G^0 | \phi_G \rangle} \quad (10)$$

Fetter-Walecka p. 61: The numerator and denominator do not separately exist because of divergencies, that properly compensate in this expression.

Time-ordered Green's function with Gell-Mann and Low Theorem:

$$\begin{aligned} & \frac{\langle \phi_G | S_\varepsilon(t_0, \infty) T [S_\varepsilon(\infty, t_0) \psi_0(1) \psi_0^\dagger(1')] | \phi_G \rangle \frac{1}{|\langle \phi_G^0 | \phi_G \rangle|^2}}{\langle \phi_G | \phi_G \rangle \frac{1}{|\langle \phi_G^0 | \phi_G \rangle|^2}} \\ &= \frac{\langle \phi_G^0 | S_\varepsilon(-\infty, t_0) S_\varepsilon(t_0, \infty) T [S_\varepsilon(\infty, t_0) \psi_0(1) \psi_0^\dagger(1')] S_\varepsilon(t_0, -\infty) | \phi_G^0 \rangle}{\langle \phi_G^0 | \underbrace{S_\varepsilon(-\infty, t_0) S_\varepsilon(t_0, -\infty)} | \phi_G^0 \rangle} \\ & \quad S_\varepsilon(-\infty, t_0) S_\varepsilon(t_0, \infty) S_\varepsilon(\infty, t_0) S_\varepsilon(t_0, -\infty) \\ &= \frac{\langle \phi_G^0 | S_\varepsilon(-\infty, \infty) T [S_\varepsilon(\infty, -\infty) \psi_0(1) \psi_0^\dagger(1')] | \phi_G^0 \rangle}{\langle \phi_G^0 | S_\varepsilon(-\infty, \infty) S_\varepsilon(\infty, -\infty) | \phi_G^0 \rangle} \end{aligned} \quad (11)$$

For a system in the ground state, we can use: $1 = |\phi_G^0\rangle\langle\phi_G^0|$.

$$G(1, 1') = -\frac{i}{\hbar} \frac{\langle\phi_G^0|S_\varepsilon(-\infty, \infty)|\phi_G^0\rangle\langle\phi_G^0|T[S_\varepsilon(\infty, -\infty)\psi_0(1)\psi_0^\dagger(1')]| \phi_G^0\rangle}{\langle\phi_G^0|S_\varepsilon(-\infty, \infty)|\phi_G^0\rangle\langle\phi_G^0|S_\varepsilon(\infty, -\infty)|\phi_G^0\rangle} \quad (12)$$

$$\boxed{G(1, 1') = -\frac{i}{\hbar} \frac{\langle\phi_G^0|T[S_\varepsilon(\infty, -\infty)\psi_0(1)\psi_0^\dagger(1')]| \phi_G^0\rangle}{\langle\phi_G^0|S_\varepsilon(\infty, -\infty)|\phi_G^0\rangle}} \quad (13)$$

... time-ordered Green's function for a system in the ground state (T=0) using the Gell-Mann and Low theorem

Interpretation:

- i) Expectation values are now calculated in the non-interacting ground state: $\langle\phi_G^0|\dots|\phi_G^0\rangle$.
- ii) $S_\varepsilon(t_0, -\infty)$ describes the evolution of the non-interacting ground state under the influence of the adiabatically switched-on interaction into the interacting ground state.

This interaction will be part of the subsequent Feynman diagram series.

- iii) $S_\varepsilon(\infty, t_0)$ accounts for the evolution of the interacting system while particles are added and removed at $1'$ and 1 , respectively.

It describes the evolution up to $t = \infty$, where the interaction is adiabatically switched off, so that $1 = |\phi_G^0\rangle\langle\phi_G^0|$ can be introduced.

- iv) In the final result, only time ordered operator products exist.